

Rebuttal for CoRL Submission #2468

We thank the Area Chair and reviewers. We respond below to R1–ZQwd, R2–jAyD, and R3–Fcwd. The new evidence directly addresses the requested progress baselines, optimization ambiguity, mask scope, component ablations, and evaluation protocol. We will revise the paper to narrow the claim to an online, tactile-supervised short-horizon contact prior that reduces pre-contact stall.

[AC, R1, R3] Is VPCE only task progress, and how does it compare with phase/hierarchical conditioning? We thank the reviewers and AC for highlighting this valuable comparison. We added phase-4 and normalized progress-20 predictors on the same held-out trajectories. VPCE obtains AUC/F1 of .963/.759, .885/.705, and .946/.735 on SWIPE/WIPE/INSERT, versus .853/.597, .743/.406, and .808/.539 for progress and .670/.000, .666/.000, and .684/.000 for phase. Thus trajectory time explains part of contact timing, but current wrist RGB and robot motion contain additional contact-specific information.

TABLE I
CONTACT PREDICTION AUC/F1 USING PHASE, PROGRESS, OR VPCE.

Task	Phase-4	Progress-20	VPCE
SWIPE	.670/.000	.853/.597	.963/.759
WIPE	.666/.000	.743/.406	.885/.705
INSERT	.684/.000	.808/.539	.946/.735

We additionally evaluated the conditioned policies under the same real-robot SWIPE protocol. Phase-DP succeeds in 12/20 trials (60%), Progress-DP in 14/20 (70%), and VPCE-DP in 28/30 (93.3%). Thus the gain is not only in contact-label prediction: replacing nominal phase/progress with the online VPCE horizon also improves policy execution. This is a controlled comparison within the same diffusion-policy family, not a reproduction of every hierarchical method; we will add Hierarchical Diffusion Policy and CompILE and avoid claiming superiority over all hierarchical policies. Unlike a discrete phase switch, VPCE continuously conditions one action generator with an online, state-dependent 15-step contact horizon.

[R1, R3] Does the teacher transfer, and how much target tactile data is required? We do not claim universal zero-shot transfer. For downstream deployment, we train target-domain teachers using approximately 60 tactile demonstrations per task. The target teachers obtain held-out AUC/F1 .912/.731, .960/.774, and .940/.725 on WIPE/SWIPE/INSERT.

[R2] Learned loss weights collapse toward the lower bound. The original intent was to emphasize sparse but control-critical contact transitions that can be underrepresented by abundant free-space samples. We agree, however, that optimizing γ and δ through the same weighted denoising objective creates a direct incentive to reduce the weight. Checkpoint inspection shows weights of only 1.000–1.005; the lower

bound prevents down-weighting below ordinary DP loss, but makes the learned component largely ineffective. We therefore replace it with the fixed schedule $w(c) = \text{clip}(1 + c, 1, 2)$, which has no trainable weighting parameters and preserves policy functionality in additional robot checks. The submitted no-weighting variant already obtains 8/10 on SWIPE, showing that the main contact-anticipation gain does not depend on learned loss weighting. We will therefore present the fixed term as a well-posed, secondary training mechanism rather than claim that weighting alone improves success.

[R2] Does Eq. (9) mask proprioception, and how is free-space motion smooth? The reviewer is correct that the paper’s notation and original post-encoder routing conflated proprioception, gripper width, and tactile features. Although input-level dropout targeted tactile keys, the post-encoder mask could zero the entire low-dimensional tail. We corrected the routing to always retain robot state and gripper width and mask only tactile/contact features. This controlled variant obtains 13/20 on SWIPE and produces more coherent free-space approach motion. Action-chunk prediction and receding-horizon execution provide additional temporal continuity. We will revise Eq. (9) to distinguish f_{prop} from f_{tac} .

[AC, R3] Which components matter, and why retain gating? We thank the Area Chair and R3 and will move the compact component ablation from the appendix into the main text. The submitted SWIPE results are full 9/10, no horizon 2/10, horizon only 0/10, no tactile masking 0/10, no gating 9/10, and no loss weighting 8/10. These results identify future-contact horizon conditioning and contact-adaptive tactile masking as the core success mechanisms, with loss weighting serving as secondary training emphasis. Because removing gating does not reduce binary success, we will downgrade it from a central contribution to an auxiliary, task-dependent modulation and retain its feature-weight trace only as diagnostic evidence.

[R2] Could operator-held props assist the policy? The reader and foam were held only to maintain a predefined height and orientation in the reachable workspace; the holder did not track the end effector, adjust the object online, or assist alignment after rollout start. All methods used the same placement, success criterion, and 60-s timeout. We expanded the original VPCE-DP evaluation to 30 trials per task: SWIPE 28/30 (93.3%, Wilson 95% CI [78.7, 98.2%]), WIPE 22/30 (73.3%, [55.6, 85.8%]), and INSERT 18/30 (60.0%, [42.3, 75.4%]). These results are consistent with the submission, but we agree that operator compliance remains a confound and will not use this protocol to claim fixed-environment robustness.

The height constraint also clarifies the method’s scope: VPCE predicts only 15 future steps and is not a long-range approach planner. Starting far outside this horizon can leave contact probability low and reintroduce stalling. We will add this limitation and investigate a height-matched rigid fixture

plus an external-view camera or coarse approach controller that first reaches the interaction region before wrist-based VPCE regulates imminent contact.

[R2, R3] Why exclude tactile input from VPCE, and how are labels and contact directions handled? VPCE intentionally excludes tactile input because its role is to anticipate contact before touch is informative and to permit tactile-free deployment; tactile is used as privileged training supervision. In the full policy, tactile feedback remains available after contact. For the current planar tasks, contact labels use the temporal change of a base-frame z -force proxy. A threshold of 0.095 is used for the source, SWIPE, and INSERT data; WIPE uses 0.0507 due to its different sensor scale. Each threshold is selected once from the training force-change distribution and then fixed. A $0.6\times-1.4\times$ threshold sweep changes the 15-step positive-label rate smoothly (SWIPE $.296 \rightarrow .217$, WIPE $.582 \rightarrow .379$, INSERT $.390 \rightarrow .290$), rather than creating labels only at one tuned value. Contacts with changing normals can instead use $\|\Delta\mathbf{F}\|_2$, local normal/shear changes, hysteresis, or learned change-point labels without changing the VPCE-policy interface.

[R1, R3] What is the main contribution relative to predictive force guidance and tactile representation learning?

We will sharpen the claim: the contribution is the identification of pre-contact stall, tactile-supervised distillation of a compact future-contact horizon into visual-proprioceptive inputs, and evidence that this signal can initiate contact even without tactile sensing at deployment. The individual gates and loss weights are not claimed as the primary novelty. Unstructured tactile encoders such as Sparsh are complementary for representing post-contact feedback; comparing them does not replace the question addressed here—how to act before tactile observations become informative. We will expand related work and state the missing Sparsh/full hierarchical policy comparisons as evaluation limitations.